

Web-Based Public Complaint Management System for Pradeshiya Sabha

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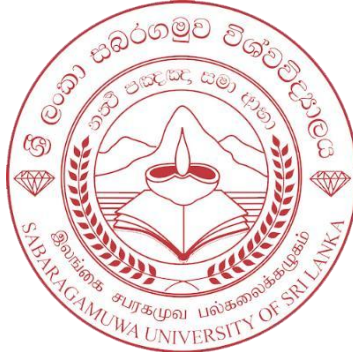
BSc. (Honors) in Computing and Information Systems

Department of Computing and Information Systems

Faculty of Computing

Sabaragamuwa University of Sri Lanka

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Report submitted in partial fulfillment of the requirement for
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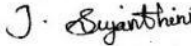
August 2026

1 DECLARATION

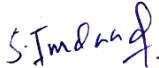
We hereby declare that the report entitled "Web-Based Public Complaint Management System for Pradeshiya Sabha" was submitted to the Department of Computing and Information Systems, Faculty of Computing, Sabaragamuwa University of Sri Lanka. The report submitted herewith of the results of our effort in totality and every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged.

Also, we hereby grant to the Sabaragamuwa University of Sri Lanka the non-exclusive right to reproduce and distribute our thesis, in whole or in part, in print, electronic, or other medium. We retain the right to use this content in whole or part in future works (such as articles or books).

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2 CERTIFICATION OF APPROVAL

I hereby declare that this report is from the student's own work and effort, and all other sources of information used have been acknowledged. This report has been submitted with my approval.



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3 ACKNOWLEDGEMENT

We extend our sincere gratitude to Prof. BTGS. Kumara, Dean of the Faculty of Computing at Sabaragamuwa University of Sri Lanka, for his invaluable guidance and support in steering us in the right direction. We are also deeply thankful to Mr. GACA Herath, Head of the Department of Computing and Information Systems, for his timely and insightful guidance.

Our heartfelt appreciation goes to Ms. A. W. T. Dulmini Lecturer (Temporary) and Coordinator of the Community Project, Department of Computing and Information Systems, Faculty of Computing, for her direction and support throughout this project. We would like to express our sincere gratitude to our internal supervisor, Mrs. P.K.D.K. Kaushalya, for her invaluable guidance, support, and encouragement throughout this project.

We also extend our thanks to the Department of Computing and Information Systems, Faculty of Computing, Sabaragamuwa University of Sri Lanka, for providing us with the necessary resources and opportunity to undertake this community project. We would also like to express our sincere thanks to Addalaichenai Pradeshiya Sabha for their trust and collaboration. Their input and cooperation have been crucial in shaping this project, and we are grateful for the opportunity to contribute to enhancing their public service delivery through digital means.

Finally, we thank our mentor, Mr. Ahamed Rujhan (Associate Software Engineer at Omobio), for his technical insights and practical advice during the development phase of the system.

4 ABSTRACT

Efficient handling of public complaints is essential for maintaining community satisfaction and transparency in local government services. Many Pradeshiya Sabhas in Sri Lanka still rely on manual or paper-based processes, which result in delayed response times, difficulty in tracking complaints, and reduced accountability. This report presents the design, development, and evaluation of a Web-Based Public Complaint Management System developed for Addalaichenai Pradeshiya Sabha.

The system enables citizens to register, submit complaints under predefined categories (Waste Management, Drainage, Road & Constructions, Streetlamp issues, and Other Issues), attach supporting evidence such as photographs and location data, track complaint status in real time, and communicate with administrators through an integrated messaging feature. An administrative dashboard allows Sabha officers to view, filter, update status, and respond to complaints efficiently. The application was built using Next.js 14, Tailwind CSS, MongoDB, and JWT-based authentication, and deployed on a cloud platform.

Evaluation results demonstrate improved accessibility, reduced paperwork, faster complaint resolution workflows, and enhanced transparency between the public and the local authority. The system is cost-effective, user-friendly, and scalable for future enhancements such as SMS/email notifications and mobile application support.

Keywords: Complaint Management System, Pradeshiya Sabha, Web Application, Next.js, E-Governance, Public Service Delivery

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Table 2.1: List of Abbreviation

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8 LIST OF ABBREVIATIONS

API	Application Programming Interface
DFD	Data Flow Diagram
ER	Entity Diagram
JWT	JSON Web Token
ODM	Object Document Mapper
UI	User Interface
UX	User Experience
VPS	Virtual Private Server
XSS	Cross-Site Scripting
HTTPS	Hypertext Transfer Protocol Secure

CHAPTER 1: INTRODUCTION

1.1 Introduction

Efficient handling of public complaints is vital for maintaining community satisfaction and transparency in local government services. In Sri Lanka, Pradeshiya Sabhas serve as the primary local authorities responsible for essential public services such as waste management, road maintenance, drainage systems, and street lighting. However, many of these institutions continue to rely on manual or paper-based complaint registration and tracking processes.

This project addresses this gap by developing a Web-Based Public Complaint Management System specifically tailored for Pradeshiya Sabha operations. The system provides a unified digital platform through which citizens can submit, track, and receive feedback on complaints, while administrative officers can efficiently manage, prioritize, and resolve issues. The solution was developed as a community project in collaboration with Addalaichenai Pradeshiya Sabha under the IS5109 – IS Project for Community module.

By digitizing the complaint lifecycle—from submission through resolution—the platform aims to increase accountability, reduce response times, improve communication between the local authority and the public, and establish a transparent record of service delivery performance.

1.2 Major goals and objectives

The primary goal of this project was to design, implement, and evaluate a secure, responsive, and user-friendly web application that digitizes the public complaint handling process for a Pradeshiya Sabha. The specific objectives were:

- To develop a secure and responsive web application for complaint registration and management.
- To provide a public interface for citizens to register, log in, and submit complaints with details such as title, description, address, location pin, and photo evidence.
- To create an administrative dashboard for Sabha officers to view, filter (by date, category, and status), and manage complaints.
- To implement a communication channel (feedback/enquiry system) between administrators and complainants.
- To ensure the system is user-friendly, accessible, cost-effective, and suitable for real-world deployment in a local government setting.

1.3 Motivation

The motivation for this project stems from the observed inefficiencies in traditional complaint handling mechanisms at the local government level. Citizens often face difficulties in submitting complaints, tracking their progress, and obtaining timely feedback. Paper-based records are prone to loss, duplication, and lack of transparency. For Sabha officers, managing a growing volume of complaints without digital tools leads to delayed resolutions and difficulty in prioritizing issues.

Digital transformation of local governance services aligns with broader national e-governance initiatives in Sri Lanka. Providing a simple, accessible web-based platform empowers citizens, strengthens accountability of public officers, and creates a foundation for data-driven decision-making regarding infrastructure and service improvements.

1.4 The scope of the completed project

The completed system covers the full complaint lifecycle for five main categories: Waste Management Issues, Drainage-Related Issues, Road & Construction, Streetlamp Not Working, and Other Issues. Key features delivered include:

- Citizen registration and secure login.
- Complaint submission with optional photo upload and location data.
- Real-time status tracking for complaints.
- Administrative dashboard with filtering and status management.
- Two-way messaging between user and administrators.
- User management capabilities for administrators.

The system is limited to web access (responsive design supports mobile browsers) and does not currently include native mobile applications, automated SMS/email notifications, or advanced analytics dashboards beyond basic filtering.

1.5 The approach and assumptions of the project

An Agile methodology was adopted, with development organized into short iterative sprints of one to two weeks. This allowed continuous feedback from the supervisor, mentor, and representatives of the Pradeshiya Sabha. The technology stack was selected for rapid development, modern performance characteristics, and low operational cost: Next.js 14 (frontend and backend API routes), Tailwind CSS, MongoDB with Mongoose, and JWT authentication.

Key assumptions included: (1) end users (citizens) possess basic digital literacy and access to smartphones or computers with internet connectivity; (2) Sabha officers have access to computers for administrative tasks; (3) the free tiers of hosting and database services would be sufficient for initial deployment and pilot usage; and (4) the five predefined complaint categories adequately cover the majority of issues handled by the Pradeshiya Sabha.

1.6 Summary of major outcomes

The project successfully delivered a fully functional web-based complaint management system that has been demonstrated to stakeholders. Major outcomes include a working public portal and administrative backend, comprehensive technical documentation (use case, ER, and DFD diagrams), successful integration of authentication and secure data handling, and positive initial feedback regarding usability and potential impact on service delivery transparency.

CHAPTER 2: **BACKGROUND**

2.1 Context

Pradeshiya Sabhas constitute the third tier of government in Sri Lanka and are responsible for a wide range of local services. Addalaichenai Pradeshiya Sabha, like many similar bodies, has historically managed public complaints through physical registers, verbal reports, or informal channels. This context creates challenges in accountability, response time measurement, and equitable service distribution.

The increasing penetration of smartphones and internet access in rural and semi-urban areas of Sri Lanka creates a favorable environment for introducing digital public service platforms. Citizens increasingly expect the convenience of online interactions that they experience with commercial services.

2.2 Problem Identification

The core problems identified through discussions with stakeholders and observation of existing processes are:

1. Delayed complaint registration and acknowledgment due to reliance on physical presence or paper forms.
2. Difficulty for citizens to track the status of previously submitted complaints.
3. Lack of a centralized digital record leading to potential loss or incomplete handling of complaints.
4. Limited two-way communication between complainants and responsible officers.
5. Absence of analytical visibility into the volume, categories, and resolution rates of complaints.

2.3 Review of Existing Solutions

Several commercial and open-source complaint management systems exist globally and within Sri Lanka. National-level platforms such as the Sri Lanka Police online complaint portal and various municipal council portals demonstrate the feasibility of digital complaint intake. However, these systems are often designed for larger institutions and may not be easily adaptable or affordable for smaller Pradeshiya Sabhas.

Generic ticketing systems (e.g., those used in customer support) offer similar workflows but lack domain-specific categorization, location mapping integration tailored to local geography, and the simplicity required for both low-literacy users and resource-constrained administrative staff. Custom development was therefore preferred to ensure precise alignment with the operational needs and budget constraints of the target Pradeshiya Sabha.

2.4 Justification for the Project

The project is justified on multiple grounds. Technically, modern web frameworks enable rapid development of secure, responsive applications at low cost. Operationally, the system directly addresses documented pain points of both citizens and officers. Economically, the

estimated annual operating cost is negligible compared with the potential gains in efficiency and citizen satisfaction. Socially, the platform promotes transparency and strengthens the relationship between the local authority and the community it serves.

Furthermore, the project contributes to the broader digital transformation agenda of local government institutions in Sri Lanka and provides a reusable model that other Pradeshiya Sabhas could adapt.

CHAPTER 3: SPECIFICATION AND DESIGN

3.1 System analysis

3.1.1 Problem Analysis

A detailed analysis of the existing manual process revealed bottlenecks at registration, assignment, status updates, and feedback stages. Stakeholders confirmed that a digital system with clear status visibility and messaging would significantly improve both internal workflow and external communication.

3.1.2 Requirement Analysis

3.1.2.1 Functional Requirements

1. The system shall allow new users to register by providing necessary details (username, email, NIC, and password).
2. Registered users and administrators shall be able to log in securely using JWT-based authentication.
3. Registered users shall be able to submit complaints under five categories with title, description, address, optional location, and optional photos.
4. Administrators shall be able to view all complaints in a dashboard and filter by category, date, and status.
5. Administrators shall be able to update complaint status (Pending, In-Progress, Resolved).
6. Both users and administrators shall be able to exchange messages regarding a specific complaint.

3.1.2.2 Non-Functional Requirements

- Usability: The interface shall be intuitive for users with basic computer skills.
- Performance: Pages should load within three seconds under normal network conditions.
- Availability: The system should aim for 99% uptime.
- Security: Passwords shall be hashed (bcrypt); data transmission shall use HTTPS; inputs shall be validated and sanitized against injection and XSS attacks.

3.2 System design

3.2.1 Data design

The data model centers on four primary collections: User, Complaint, Message, and Analytic_logs. Users have roles (public or admin). Complaints reference the submitting user and store category, description, address, photo references, status, and timestamps. Messages link senders, receivers, and the related complaint. Analytic logs record status changes for audit purposes. An Entity-Relationship diagram was prepared to formalize these relationships.

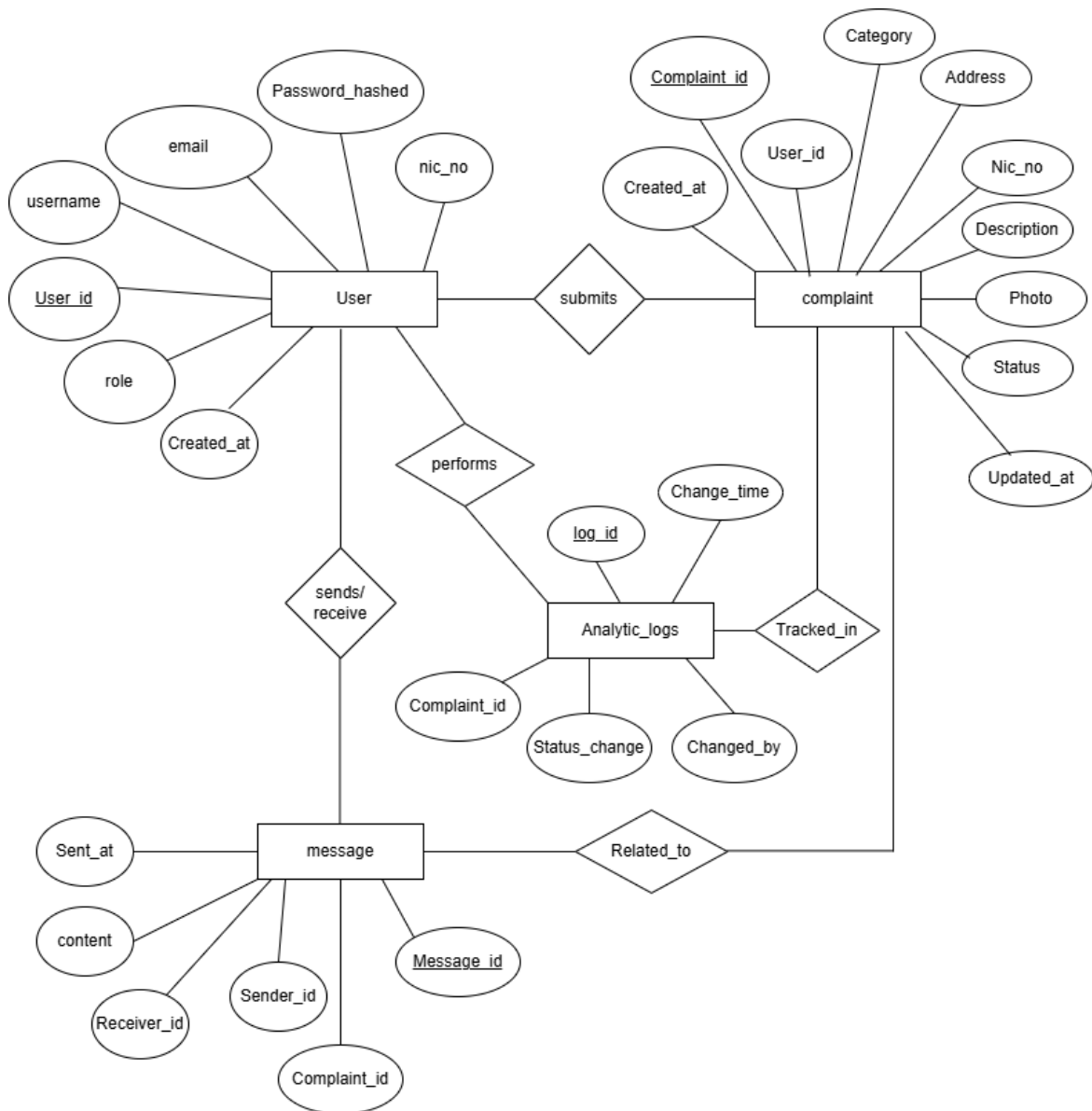


Figure 3.2.1.1. ER Diagram

3.2.2 Process design

Process design was captured through Use Case Diagrams (showing Public, Registered Complainant, Admin, and external Google Maps API actors), Data Flow Diagrams at Level-0 and Level-1, and supporting UML activity and sequence representations for key flows such as complaint submission and status update. These diagrams guided the modular implementation of registration, authentication, complaint CRUD operations, filtering, and messaging.

Use Case Diagram

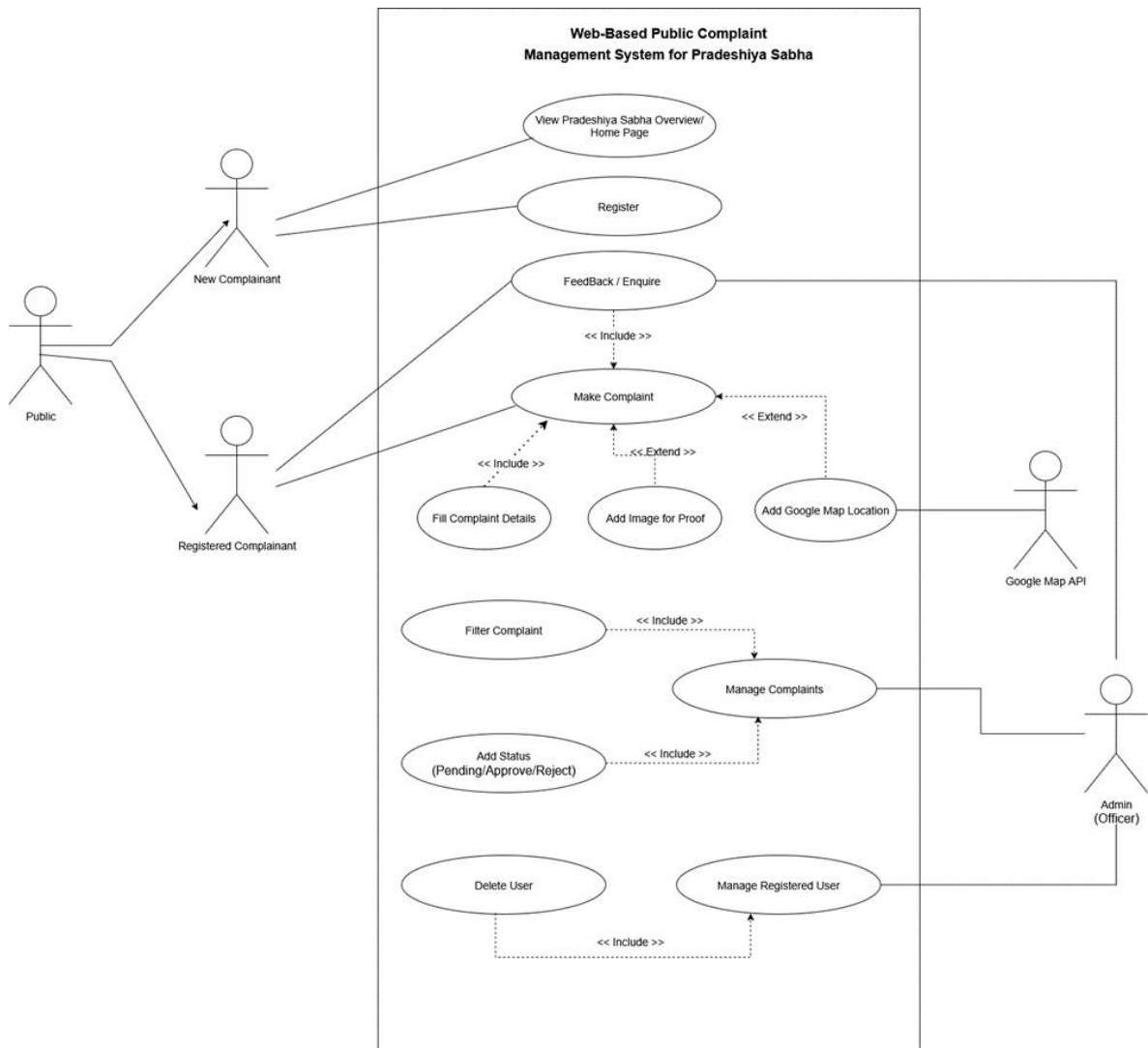


Figure 3.2.2.1. Use Case Diagram

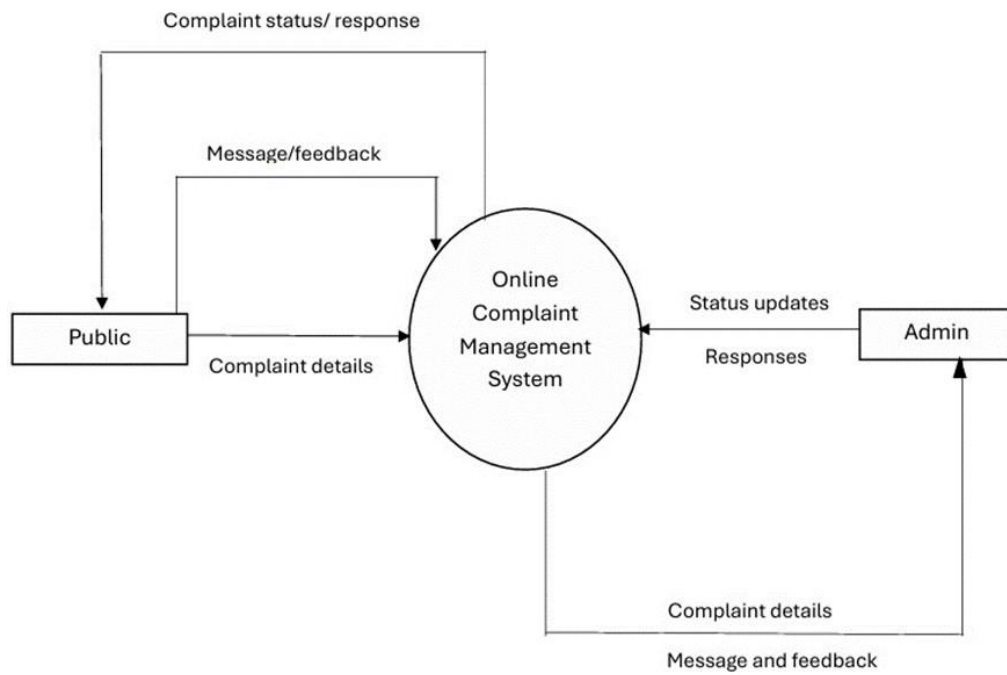
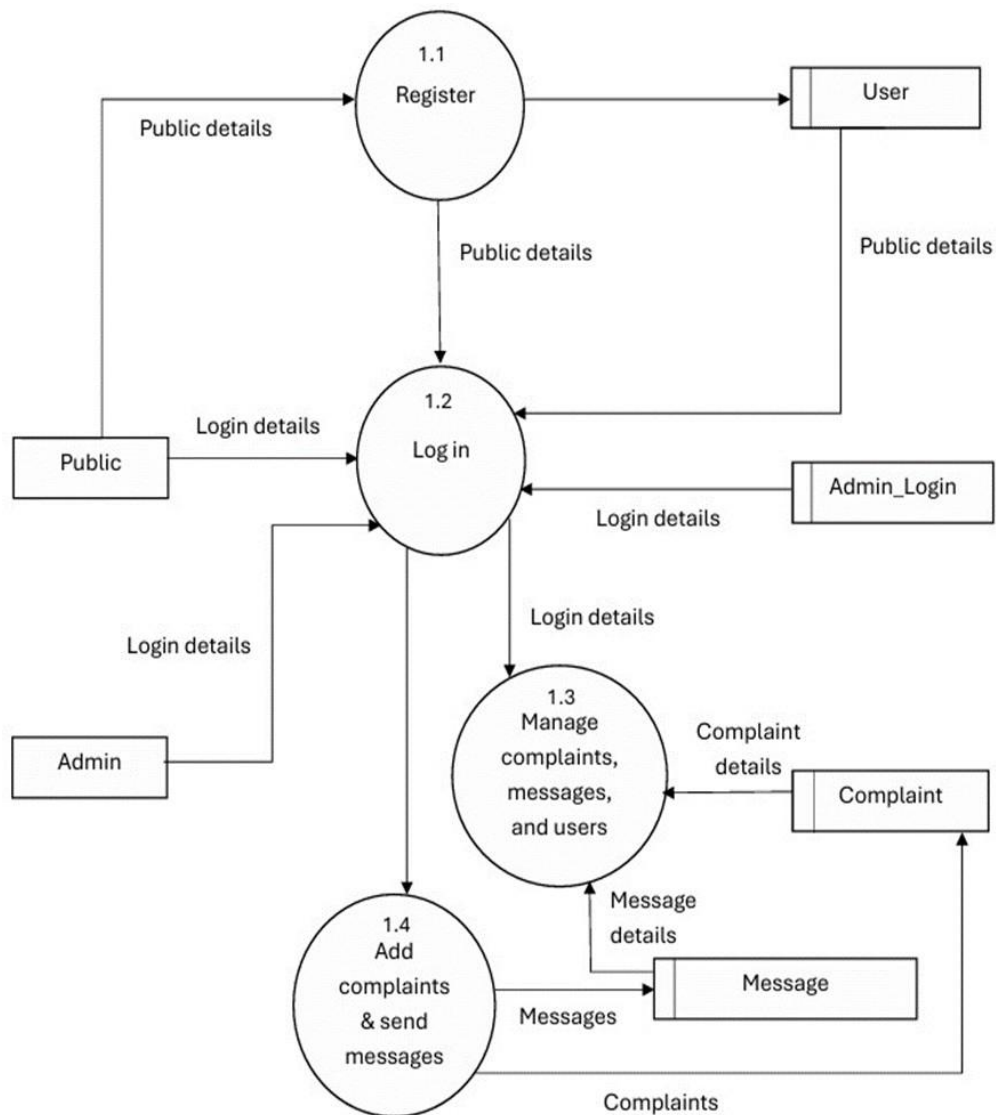
Data Flow Diagram (DFD)**DFD Level 0**

Figure 3.2.2.2. DFD Level 0

DFD Level 1*Figure 3.2.2.3. DFD Level 1***3.2.3 User interface design**

The user interface was designed with a clean, responsive layout using Tailwind CSS. Public pages emphasize simplicity and clear calls to action. The administrative dashboard prioritizes information density with filter controls and status indicators. Wireframes and high-fidelity mockups were produced in Figma prior to implementation to validate usability with stakeholders.

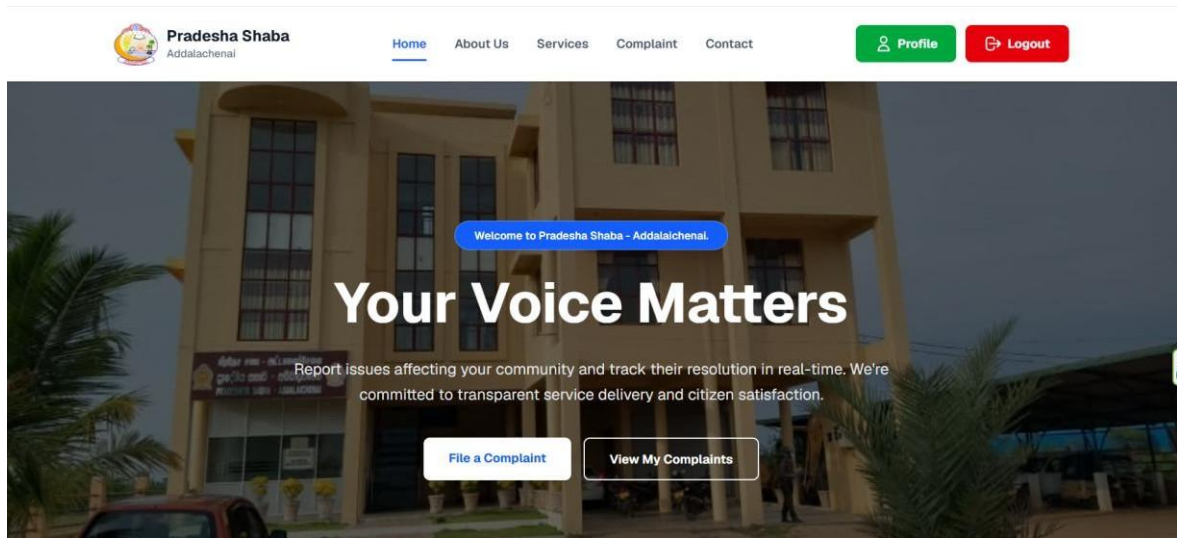


Figure 3.2.3.1. System User Interface Design (1)

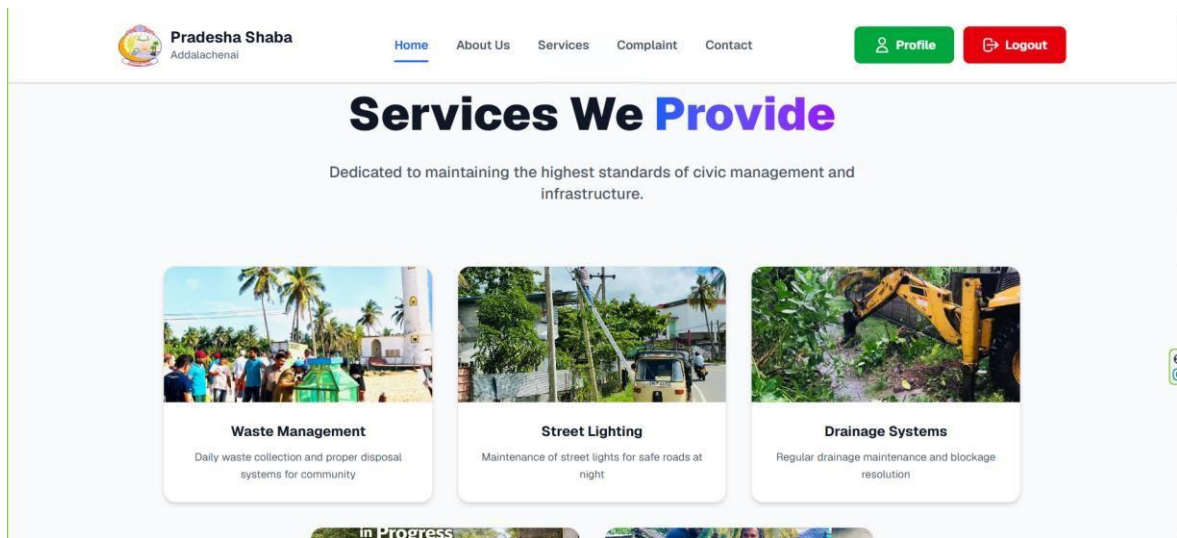


Figure 3.2.3.1. System User Interface Design (2)

CHAPTER 4: IMPLEMENTATION

4.1 Software and Hardware Requirements

Development was performed on standard personal computers (minimum 4 GB RAM, Core i3 or equivalent) running Windows, macOS, or Linux. Production deployment targets cloud infrastructure (Vercel for the Next.js application and MongoDB Atlas for the database). Software tools included Visual Studio Code, Git, Figma, and Draw.io. The core technology stack comprised Next.js 14, Tailwind CSS, JWT, MongoDB with Mongoose, and bcrypt for password hashing.

4.2 Illustration of implementing an algorithm and data structure

Complaint filtering and status management rely on efficient MongoDB queries with indexes on category, status, and created_at fields. Authentication implements the standard JWT flow: credentials are verified, a signed token containing user ID and role is issued, and protected API routes validate the token on each request. Password storage uses bcrypt with an appropriate work factor. File uploads for complaint photos are handled securely with validation of file type and size before storage.

4.3 Difficulties involving existing software.

Integration of location services required careful handling of the Google Maps API key and client-side map component loading. Ensuring consistent session behavior across Next.js server and client components necessitated attention to cookie and header configuration for JWT transmission.

4.4 Lack of appropriate supporting software

No major gaps in supporting software were encountered. The chosen open-source stack provided all required capabilities. Minor challenges arose around configuring free-tier hosting limits and ensuring cold-start performance remained acceptable for a serverless deployment model.

4.5 Over-ambitious project aims

Initial discussions considered SMS notifications and a native mobile application. These were deferred to future work to keep the core system deliverable within the available time and resource constraints. The implemented web application fully satisfies the primary objectives of digital complaint intake, tracking, and two-way communication.

CHAPTER 5: RESULTS AND EVALUATION

5.1 The comparison of experimental results with expected values

Functional testing confirmed that all specified requirements were met. User registration, login, complaint submission (including photo and location), status filtering, status updates, and messaging operated as designed. Page load times under normal conditions remained within the three-second target. Security tests verified that unauthorized access to admin routes was blocked and that passwords were stored only in hashed form.

5.2 Description of the interrelationship of the experimental results

Successful end-to-end workflows demonstrated the correct interrelationship between modules: a newly registered user could submit a complaint that immediately appeared on the admin dashboard; status changes made by an administrator became visible to the complainant; and messages exchanged on a complaint thread were correctly associated and displayed to both parties.

5.3 Analyze and state the achieved accuracy

The system achieved high functional accuracy against the requirements specification. Category classification, status transitions, and role-based access control behaved consistently across repeated test cases. Input validation correctly rejected malformed or incomplete submissions.

5.4 Analyze and state implications or limitations.

The principal implication is that a low-cost, modern web stack is sufficient to deliver a practical e-governance tool for a Pradeshiya Sabha. Limitations include dependence on internet connectivity, the absence of offline capability, and the current restriction to browser-based access. Photo storage and bandwidth considerations may require monitoring as usage scales. The system currently relies on manual status updates by officers; automated escalation rules remain future work.

CHAPTER 6: FUTURE WORK

6.1 Gaps of the project

Identified gaps include the lack of automated notifications (SMS/email), absence of advanced analytics and reporting dashboards, no native mobile applications, limited multi-language support (currently English-oriented), and the need for more extensive user acceptance testing with a larger sample of citizens and officers under real operational conditions.

6.2 Proposal for enhancement or re-design

Recommended enhancements include:

- Integration of SMS and email notification gateways.
- Development of a progressive web app or native mobile clients.
- Addition of role-based officer assignment and workload balancing.
- Incorporation of simple analytics (complaint volume trends, average resolution time by category).
- Potential integration with other local government information systems. A redesign of the notification and escalation layer would further improve responsiveness.

CHAPTER 7: CONCLUSION

7.1 The importance of the result

The completed Web-Based Public Complaint Management System demonstrates that a practical, transparent, and affordable digital platform can be delivered for a Pradeshiya Sabha using contemporary open-source technologies. The system directly addresses long-standing inefficiencies in complaint handling and provides a foundation for improved citizen engagement and administrative accountability.

7.2 Validity of the result

The validity of the outcome is supported by successful functional testing against the original requirements, positive stakeholder feedback during demonstrations, and alignment of the delivered features with the documented needs of Addalaichenai Pradeshiya Sabha. The technical architecture follows established best practices for security and maintainability.

7.3 Gaps and limitations of the findings

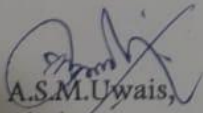
While the core system is complete and operational, real-world performance metrics under sustained public usage, long-term maintenance requirements, and quantitative measures of citizen satisfaction remain to be collected after formal deployment. These limitations do not diminish the value of the delivered solution but indicate directions for continued evaluation and improvement.

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10 APPENDIX

Appendix A: Letter from the organization

	பிரதேச சபை - அட்டாளைச்சனை ප්‍රදේශීය සභාව - අඩ්ඩාලච්චේන Pradeshiya Sabha - Addalaichenai	
Main Street, Addalaichenai	E-mail: adpslg24@gmail.com	Tel: 067-2277375 / Fax: 067-2278600
Your No:	My No: AD/PS/PA/01/GEN/2026	Date: 2026.08.21
TO WHOM IT MAY CONCERN This is to certify that Community Project Group 17 of the Faculty of Computing, Sabaragamuwa University of Sri Lanka, has successfully developed and implemented a Web-Based Complaint Management System for the Addalaichenai Pradeshiya Sabha as part of their IS5109 - Community Project. The completed system was duly demonstrated and tested and was found to be functioning satisfactorily in accordance with the project requirements. The project team has successfully completed the project and formally submitted and handed over the completed system to the Addalaichenai Pradeshiya Sabha for its intended use. The Addalaichenai Pradeshiya Sabha appreciates the valuable contribution made by the project team towards improving public complaint management, administrative efficiency, and service delivery through the use of digital technology. This letter is issued as official confirmation of the successful completion, submission, and handover of the project. Yours faithfully,  A.S.M. Uwais, Chairman AS.MOHAMEDU UWAIS Chairman Pradeshiya Sabha Addalaichenai		